

Junaid Muhammad

M.S. (Info. & Comm. Engg.) | Focused in AI and Computer Vision

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Jiho-ro, Jisan-dong, Dong-gu, Gwangju - 61445, South Korea

OBJECTIVE

Master's graduate specializing in artificial intelligence for computer vision and medical image analysis, with hands-on experience in deep learning models such as CNNs, diffusion models, and transformer-based architectures, as well as clinically validated dataset curation. Research interests focus on AI-driven solutions for real-world problems across computer vision and healthcare applications.

EXPERIENCE

• Chosun University

Graduate Research Assistant

Mar 2024 - Feb 2026

Gwangju, South Korea

- Conducted research in medical image processing and image restoration.
- Lead the curation of a clinically validated musculoskeletal multimodal dataset in collaboration with medical professionals, and developed a survival analysis framework for predicting fracture risk.
- Developed Residual-CNN based deep learning frameworks for bone and hip fracture detection using pytorch.
- Worked on transformer and diffusion based architecture for image processing.

• University of Swat

Administrative Assistant

Jan 2020 - Feb 2024

Swat, Pakistan

- Academic and administrative tasks covering student admissions, records management, examination and results processing, transcript, degree and certificate issuance.

EDUCATION

• Chosun University

M.S. Information and Communication Engineering

Mar 2024 - Feb 2026

Gwangju, South Korea

- GPA: 4.11/4.50; Percentage: 95.32; Thesis: **A Study on Hierarchical Pyramid Context Refinement with Supervised Contrastive Learning for Hip Fracture Classification from Radiographs**; Thesis word count: 12156

• Iqra National University

B.S. Telecommunication

Sep 2012 - Feb 2017

Peshawar, Pakistan

- GPA: 2.96/4.00; Percentage: 75.79; Thesis project: DTMF controlled IP based live broadcasting robot with night vision - Thesis and end-to-end hardware and system development

CONFERENCE PAPERS PRESENTATIONS

- [1.] Junaid Muhammad, Bumshik Lee. (2025). **"Dual Attention Transformer for Image Super Resolution"**, Oral presentation, 01-03 May, 2025, Korean Institute of Smart Media and E-Commerce (KISM) Spring Academic Conference, Seoul, South Korea.
- [2.] Junaid Muhammad, Bumshik Lee. (2025). **"Bone Abnormality Classification with Supervised Contrastive Learning on MURA"**, Oral presentation, 06-09 November, 2025, The 26th International Symposium on Advanced Intelligent Systems, Cheongju, South Korea.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, R=IN REVISION, T=THESIS

- [R.1] Junaid Muhammad, Bumshik Lee. (2026). **HPCRNNet: A Hierarchical Pyramid Context Refinement Framework for Interpretable Hip Fracture Analysis**. *Applied Soft Computing (ASOC)*, In revision, 2026.

SKILLS

- **Programming Languages:** Python, C++
- **Data Science & Machine Learning:** PyTorch, TensorFlow, Huggingface, timm, OpenCV, NumPy, Scikit-learn, CUDA, Jupyter Notebook, PyCharm
- **Specialized Area:** Computer Vision, Image Processing, Deep Learning, Medical Image Analysis, Medical Imaging Dataset Curation, Transformers, Diffusion Models, Convolutional Neural Networks, Residual Learning
- **Research Skills:** End-to-End Research Workflow, Scientific Writing, Benchmarking, Strong Written & Communication Skills, Data Visualization

HONORS AND AWARDS

• Chosun University Foreign Excellence Scholarship

Chosun University, Gwangju, South Korea

Mar 2024 - Feb 2026

CERTIFICATIONS

- **IELTS Academic:** Overall Score 7.0

Mar 2026

ADDITIONAL INFORMATION

Languages: English (Fluent), Urdu (Native), Pashto (Native), Korean (Basic Proficiency)

Interests: Hiking, Sports, Reading

REFERENCES

1. **Dr. Arif Ullah**

Research Fellow, Queen's University, Belfast, UK

Email: a.ullah@qub.ac.uk

[GitHub](#)

Relationship: Former Assistant Professor, Dept. of Information and Communication Engineering, Chosun University

2. **Dr. Fawad**

Postdoctoral Researcher, Korea Institute of Energy Technology (KENTECH), South Korea

Email: Fawad1@kentech.ac.kr

[Google Scholar](#)

Relationship: Colleague at ENVI Lab, KENTECH, South Korea - Former Postdoctoral researcher at Chosun University